All of Us Scientific Roadmap Refresh — Executive Summary

2026 Science Committee Working Group · Co-leads: Jordan Smoller & Melissa Haendel · Draft 2026-10-14

THE CHARGE

The Science Committee asked this working group to refresh the All of Us Scientific Roadmap against the six Roadmap 2.0 questions: what has changed since 2023, the largest data gaps, reproducibility expectations, the new tools researchers need, which exposures and outcomes to capture, and — centrally — **whether the program's eight priority focus areas remain the right ones.** This charge is uniquely structural: only this group can adjust the framework within which the other four 2026 working groups operate.

HOW WE ARRIVED AT THESE FINDINGS

Every recommendation is traceable to evidence. We classified the program's research output — **1,653 publications and 20,165 active research projects** (from a live 32,922-workspace production pull, 2026-10-14) — against a 19-theme taxonomy and mapped it onto the eight focus areas, with pre- and post-2023 trends. We assessed alignment to NIH priorities by extracting **166 priority statements from 29 NIH strategic plans** (the NIH-wide plan plus 21 Institutes and Centers). We benchmarked All of Us against the **four leading peer biobanks** (UK Biobank, FinnGen, the Million Veteran Program, and China Kadoorie — complete publication sets, the same method applied uniformly). And we scanned the leading edge of the field — roughly **322,000 preprints and 460,000 NIH grants** (2021–2026) — for emerging opportunities the program has not yet captured.

HEADLINE FINDINGS

- **The eight focus areas remain sound** — but three bodies of work have no clear home in the framework: data science and AI methods (now 18% of all publications), wearable and remote-monitoring research, and multi-omics (today subsumed under "Genetics & Biology").
- **All of Us aligns most strongly with NHGRI, NIMHD, and NLM.** The program is the clearest delivery vehicle for NHGRI's genomics goals, embodies NIMHD's health-equity mandate (its Goal 7 reads as a description of All of Us), and is the *only* program named explicitly in an NIH Institute strategic plan (NLM, on methods and informatics).
- **The clearest unmet demand is in environment, imaging, and multi-omics.** Environmental exposures (NIEHS) show the largest gap between Institute priority and current output; imaging (NCI/NEI/NIBIB) is a question of data supply rather than researcher demand; and multi-omics demand (NIDDK/NHLBI) should be met largely by the planned 2026 proteomic/metabolomic releases.
- **Against the leading peer biobanks, All of Us has a distinctive identity worth protecting.** It is comparatively under-represented in rare disease, multi-omics, and imaging, but leads its peers decisively in ethical/social research, health-equity science, pharmacogenomics, and analytic methods. Distinctiveness — not raw publication growth — is the durable signal.
- **The field's leading edge points to a clear, feasible opportunity.** The fastest-rising methods All of Us could act on today are **genomic foundation models** and **large-language-model / agentic AI** approaches, where the program is currently almost absent (under 0.3% of its work; LLM-in-medicine publishing has grown roughly 47-fold since 2021), alongside adopting the updated MASLD fatty-liver disease definition. These map directly onto the proposed new focus area and the Workbench AI assistant. Trends that require data the program does not collect (single-cell/spatial assays, AI protein-structure prediction) are deliberately out of scope.
- **The Researcher Workbench leads on data and compute but trails on three remediable capabilities.** Benchmarked against five comparator platforms (N3C, Terra/AnVIL, BioData Catalyst, UK Biobank RAP, Kids First; every claim source-verified), the Workbench is strongest on data assets and AI-grade compute, but lacks (1) a registry for sharing reusable tools and workflows that every comparator offers, (2) a governed in-environment AI assistant (today only researcher-supplied tools exist), and (3) portable, interoperable analysis workflows. The highest-value builds are a governed in-environment AI assistant — permissible today under the program's existing cloud agreements — a reusable-tools registry, and multimodal-data templates. *An unreleased 2026 ACT-IAC/All of Us AI/ML white paper, developed independently from stakeholder interviews, reaches the same conclusions — recommending a governed in-environment AI research assistant with agent-ready metadata services, a reusable-research-asset registry with graduated tool certification, and portable workflows — corroborating this data-driven gap analysis.*
- **There is a clear basis to position All of Us as shared NIH-wide infrastructure.** Five federal strategic plans articulate explicit demand for pan-NIH data infrastructure, giving an empirical foundation for that framing.
- **Ancillary (partnered) studies are the program's leading edge — and the publication record cannot yet show them.** New partnered studies add modalities on top of the core protocol: *Exploring the Mind* (NIMH — objective cognitive/behavioral data), *Eyes on Health* (NEI/NIBIB — retinal imaging/oculomics), *Nutrition for Precision Health* (Common Fund — individualized diet response), and *Environmental Health & Exposomics* (NIEHS). Exploring the Mind has **zero publications or preprints anywhere** (verified against Europe PMC), yet already appears in **32 workspaces — 27 of them invisible in the pre-refresh workspace data**; cognition/mental-health work is now 13% of all workspaces. Workspace activity leads publications by a year or more, so the roadmap must score these studies on their stated goals and live workspace intent, not on a literature that cannot yet exist — which also argues for pulling live workspace data rather than a cached snapshot.
- **The 2026 committee meeting series surfaced concrete new directions worth building into the refresh.** Four stand out: a **driver-project model** (the Women's Health Driver Project showed how program biostatistical support yields exemplar papers — a template the committee framed for *rare-disease and intersectional* analyses, directly attacking the rare-disease peer-gap); **generalizability tooling** (a Driver Project and released `waou` R package for weighting All of Us estimates to the US population); a live **multi-omics program** (RNA-seq + Olink proteomics +

long-read on ~10,000 diverse participants, scaling toward ~50,000); and a publicly posted **external-biospecimen-request mechanism** (academic/nonprofit/industry) that pairs with a partner-contribution model. The committee also raised an **internal All of Us foundation model hosted securely with no data egress**, guided by a proposed "super user group" — which maps directly onto this analysis's two top AI recommendations (the genomic-foundation-model white-space and the governed in-environment AI assistant). A **November 2, 2026 environmental-dataset release** (geocoded residential history + linked social/environmental variables) is the near-term milestone closing the environmental peer-gap. (Details in `aou-roadmap-refresh/committee_2026_alignment.md`.)

- **Half of the 2023 priorities are already realized.** Four are now established in the research base (AI, breakthrough-therapeutic studies, wearables, and integrated gene-environment-social-determinant risk); four remain aspirational pending upstream investment (multi-omics, imaging, microbiome, environment).

TOP RECOMMENDATIONS

Recommendation	Score	Related need area(s)
Add a ninth focus area, "**Data, AI & Methods**" — consolidating the data-science and AI methods (18% of output), wearable research, and the foundation-model surge that currently have no home in the framework.	3.71	Scientific Roadmap (framework)
Rename FA8 (currently "Genetics & Biology") to "**Genetics, Multi-omics & Biology**" and commit to a proteomics-release roadmap (Olink/SomaScan scale).	3.38	Scientific Roadmap; Genomics & Multi-omics
Bundle focus-area reframes: FA4 ("Return of Results on Individual & Population Health") toward ROR implementation; FA2 ("Maternal & Child Health") to "**Maternal Health**"; add explicit FA6 ("Environment") environmental-exposure linkage commitment.	3.00	Scientific Roadmap
Workbench **AI co-scientist** (cohort-builder + code-generator) — the highest-scoring recommendation across all areas.	3.88	AI Strategies
A shared **bias-in-translation framework** (normative + imaging + AI) — a single All of Us bias-reporting standard.	3.75	Translational · Imaging · AI (coordinated)
Plain-language **distinctive-evidence narrative** + annual All of Us Impact Report (ELSI 24×, SDOH 5×, PGx 4.3×, methods 3×).	3.60	Scientific Communications

Score — mean (of 4) on the Committee's 8-criterion prioritization rubric. **Need areas** correspond to topics already recognized as program priorities (the 2026 working groups); they overlap and are meant to be coordinated across groups, not actioned or voted by any single group.

For reference — the program's current 8 priority focus areas: (1) Prevalent Common & Rare Conditions · (2) Maternal & Child Health · (3) Healthy Aging & Resilience · (4) Return of Results on Individual & Population Health · (5) Lifestyle, Substance & Behavioral Health · (6) Environment · (7) Health Equity · (8) Genetics & Biology.

NOTES ON INTERPRETATION

- Peer comparisons are most reliable as *relative emphasis* (each program's mix of topics), not head-to-head growth rates; the same identification method was applied to every program.
- All of Us's research base before 2023 is small (157 publications), so recent percentage "growth" in established areas mainly reflects a maturing program rather than outpacing the field.
- Minor: a few topic-level outliers (e.g., an elevated dermatology share) reflect a small number of prolific groups rather than strategic emphasis and are not load-bearing for these recommendations.

Full evidence base, per-IC briefs, and machine-readable companions at `github.com/mellybelly/all-of-us`